

**BEFORE THE PUBLIC UTILITIES COMMISSION
OF THE STATE OF CALIFORNIA**

Application of Southern California Edison
Company (U-338E) for Approval of Large
Power Dynamic Pricing Rate.

Application 24-06-014
(Filed September 30, 2024)

And Related Matter.

Application 24-12-008

OPENING BRIEF OF THE VEHICLE-GRID INTEGRATION COUNCIL

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Pursuant to Rule 13.1 of the Rules of Practice and Procedure of the California Public Utilities Commission (“Commission”) and the *Administrative Law Judge’s Ruling Modifying The Proceeding Schedule Including the Addition of Supplemental Testimony* issued on March 25, 2026, the Vehicle-Grid Integration Council (“VGIC”) hereby submits this Brief on the consolidated *Application of Southern California Edison Company (U 338-E) for Approval of Large Power Dynamic Pricing Rate*.

I. INTRODUCTION.

VGIC appreciates the opportunity to comment on Southern California Edison Company's ("SCE") dynamic rate design. The record in this proceeding demonstrates that SCE’s dynamic rate developed for Load Management Standard compliance should include dynamic export credits, especially for bidirectional electric vehicle charging systems (i.e., non-NEM and non-NBT exporting systems). Dynamic export credits are consistent with the Commission's clear, repeated direction that distributed energy resource (“DER”) exports: (a) be compensated based on unscaled marginal costs; (b) can be implemented using SCE's proposed dynamic rate framework; and (c) will enable customers to provide valuable grid services when those services are most needed. **In an era of outright ratepayer rage over electricity affordability, the Commission must not**

fumble an immediate opportunity to unlock grid-edge energy exports that, by definition, are provided during the most valuable hours.

When SCE developed its proposal, it anticipated that customers with bidirectional EV charging systems, which are non-NEM and non-NBT systems, would have access to export credits through the Vehicle-Grid Resource Proposal ("VGRP") in Application ("A.") 24-03-019. Following the Commission's decision not to adopt VGRP, that assumption no longer holds. As a result, customers with bidirectional EV charging systems now lack a durable pathway to provide compensated exports to the grid, despite broad agreement in this proceeding that dynamic export compensation should be based on unscaled marginal costs.

Providing a dynamic export rate would align SCE's proposal with the Commission's marginal-cost pricing principles while unlocking substantial reliability and affordability benefits from California's rapidly growing fleet of EVs, a growing share of which offer bidirectional charging features. Conversely, failing to provide a pathway for exports will encourage customers and technology providers to optimize these systems primarily for behind-the-meter applications, limiting their ability to support the electric grid and significantly reducing the broader benefits these resources can provide. For these reasons, the Commission should require SCE to include dynamic export credits for bidirectional EV charging systems (as non-NEM, non-NBT systems) in the implementation of its dynamic rate.

II. THE COMMISSION SHOULD REQUIRE DYNAMIC EXPORT CREDITS BASED ON UNSCALED MARGINAL COSTS.

A. The record demonstrates broad agreement that dynamic export credits should be based on unscaled marginal costs.

The record reflects broad agreement that export credits be included in a dynamic rate based on unscaled marginal costs consistent with D.25-08-049. The parties addressing export credits,

including Cal Advocates, SEIA, and VGIC, all share support for a dynamic export rate based on unscaled marginal costs.¹ This is in line with the Commission’s underlying Demand Flexibility Design Principles, which state that **demand flexibility tariffs should provide marginal cost-based compensation for exports.**² As discussed by SEIA, a dynamic export rate should be adopted because “Export rates that are closely linked to actual market prices and loads will provide the most economically efficient price signals.”³ In other proceedings, Cal Advocates has proposed that the Commission should “expand existing dynamic rate pilots [to include exports] and continue to test export credits based on a dynamic price signal,” since these dynamic rate frameworks can most accurately compensate for the value of exports.⁴

Additionally, Cal Advocates has highlighted a concern that a dynamic transmission rate may include recovery of non-marginal costs.⁵ However, VGIC believes that if SCE’s dynamic transmission rate includes non-marginal cost components, *those components should be removed from the export rate*, such that all export components are based solely on unscaled marginal costs, constituting a defensible, durable approach. VGIC believes that this is technically feasible, since SCE has shared the ability to separate scaled from unscaled factors and separate out marginal from non-marginal cost components.

B. SCE has confirmed that dynamic export credits can be implemented.

¹ CA-04 at p.21, lines 8-9; SEIA-01 at p.14, lines 23-27 – p.15, lines 1-6; VGIC-01 at p.15, lines 16-22 – p.16, lines 1-3.

² SEIA-01 at p.14, lines 12-15.

³ SEIA-01 at p.14, lines 15-16.

⁴ Cal Advocates comments at 4

<https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M582/K489/582489355.PDF>

⁵ CA-04 at p.22, lines 3-20.

The record contains no evidence that technical barriers prevent the implementation of dynamic export credits. In fact, **SCE itself has, in no uncertain terms, confirmed that “SCE *can* offer dynamic hourly export rates following the dynamic rates design methods as detailed in A.24-12-008,”⁶ and has outlined that SCE would be able to develop a rate, “consist[ing] of non-EPMC-scaled SCE distribution and generation capacity marginal cost rates, CAISO day-ahead market prices, and hourly transmission prices, as ordered in Decision 25-08-049.”⁷**

Notably, Pacific Gas and Electric (PG&E) has also shown that dynamic export compensation rates can be developed and implemented. PG&E currently offers export compensation in its dynamic rate pilot *and* proposes to extend dynamic rate export compensation in its long-term dynamic rate (with the subscription component becoming optional in the long-term proposal).⁸ Given the already proven ability of utilities to implement dynamic export rates, and SCE’s explicit confirmation that a dynamic export compensation rate compliant with D.25-08-049 can be feasibly implemented in its service territory, the Commission should require SCE to adopt dynamic export compensation based on unscaled marginal costs for which bidirectional EV charging systems (and likely other DERs) are eligible for credits.

III. THE FACTUAL CIRCUMSTANCES UNDERLYING SCE'S PROPOSAL HAVE MATERIALLY CHANGED.

SCE's original proposal did not include dynamic export compensation because it anticipated that customers with bidirectional EV charging systems would instead receive export

⁶ DATA REQUEST SET VGIC - SCE - 02

⁷ DATA REQUEST SET VGIC - SCE – 02 question 1.A

⁸ VGIC-01 at p.23, lines 14-20, citing PG&E, Pacific Gas and Electric Company 2023 General Rate Case Phase II Supplemental Testimony Exhibit (PG&E-5) Real-Time Pricing Proposal, served to A.24-09-014 on October 29,2025, at p.2-41, lines 1-7.

compensation through the Vehicle-Grid Resource Proposal ("VGRP") in Application 24-03-019. Under that framework, SCE was concerned that having both VGRP and a dynamic export rate would “lead to customer confusion, settlement and billing complexities, and potentially promote inequities among participating EV customers.”⁹

Since SCE filed this Application and submitted testimony outlining its proposed dynamic rate, the Commission declined to adopt VGRP. As a result, the fundamental assumption underlying SCE's export compensation proposal no longer exists. Unlike customers with solar, bidirectional EV charging customers in SCE's service territory now lack any export compensation pathway beyond summer 2027.¹⁰

In A.24-03-019, the Commission rejected VGRP because of concerns from Cal Advocates, namely concerns around “the accuracy and cost-effectiveness of ACC-based, versus real-time marginal-cost-based, credits in reflecting marginal costs.”¹¹ Those concerns do not apply to the proposal advanced here. Rather than relying on ACC values, VGIC and parties propose dynamic export compensation based on unscaled marginal costs consistent with D.25-08-049. For this reason, while Cal Advocates opposed VGRP, they can support dynamic export pricing, stating “Cal Advocates supports the Commission adopting a dynamic export rate in this proceeding only if the export rate is based only on unscaled marginal costs.”¹²

⁹ SCE-03 at p.41, lines 20-24.

¹⁰ See VGIC-01 at p.22, lines 19-21. Outside of rates, there are very limited opportunities for bidirectional charging systems to provide value to the grid through exports in SCE's service territory. The only currently available programs are the Emergency Load Reduction Program (“ELRP”), which is scheduled to sunset after 2027, and the Demand Side Grid Support Program (“DSGS”), which may not continue beyond this year. As a result, absent further Commission action, bidirectional charging systems will effectively have no pathway to provide grid services after 2027.

¹¹ D.26-04-033 at 49.

¹² CA-04 at p.21, lines 8-9.

IV. DYNAMIC EXPORT COMPENSATION WILL UNLOCK SUBSTANTIAL GRID BENEFITS.

DERs can provide tremendous value to the grid through exports, especially from resources that have not traditionally participated in Net Energy Metering (“NEM”) or Net Billing Tariff (“NBT”) tariffs. EVs are the largest source of energy storage in California, with the California Energy Commission (“CEC”) estimating that there are already 18.5 GW of aggregate vehicle battery capacity statewide, surpassing California’s installed stationary storage capacity.¹³ Every day, additional vehicles are sold in California with grid-parallel bidirectional charging features, including 24 models on the market today from 14 automotive brands, with four additional models from four more automakers announced for upcoming release.¹⁴ For SCE specifically, as of the end of 2025, VGIC estimated that there were already over 70,000 vehicles with bidirectional charging features in its service territory.¹⁵

These resources can provide significant grid value. The Commission's own Integrated Resource Planning assumptions indicate bidirectional EV charging could provide more than 13,500 MW of additional capacity by 2045 beyond the contributions from managed charging alone.¹⁶ The CEC’s report on bidirectional charging estimates that bidirectional discharge will be the largest source of customer load flexibility and grid value, providing 5 GW of residential peak

¹³ California Energy Commission. *A Roadmap to Unlocking the Benefits of Bidirectional Charging*. March 2026. Page 2. <https://efiling.energy.ca.gov/GetDocument.aspx?tn=268952&DocumentContentId=106145>

¹⁴ VGIC-01 at Table 1 (p.8-13) and p.14, lines 3-7.

¹⁵ VGIC-01 at p.13, lines 3-4.

¹⁶ VGIC-01 at p.5, lines 3-14, citing California Public Utilities Commission (“CPUC”), Draft Inputs & Assumptions, 2024 – 2026 Integrated Resource Planning (IRP) at p.111. https://www.cpuc.ca.gov/-/media/cpuc-website/divisions/energydivision/documents/integrated-resource-plan-and-long-term-procurement-plan-irp-ltp/2024-2026-irp-cycle-eventsand-materials/2025_draft_inputs_and_assumptions_doc_20250220.pdf

load reduction and approximately three times the value of unidirectional managed charging.¹⁷ These projections demonstrate that establishing a pathway for bidirectional EV charging exports is essential to realizing the reliability and affordability benefits anticipated in the Commission's planning processes.

V. **THE COMMISSION MUST ENABLE GRID-INTERACTIVE ACTIVITIES THAT ENCOURAGE CUSTOMERS TO CONTRIBUTE TO THE GRID, RATHER THAN ENCOURAGE GRID DEFECTION.**

Given the untapped potential of bidirectional EV charging, including the sheer energy capacity available in each vehicle, customers **will** seek safe ways to use these assets to their benefit. It is therefore critical to ratepayer interests that the Commission establish pathways that enable and encourage these resources to contribute to grid needs. SCE has already demonstrated significant customer interest in “V2G energy exports,” with 81% of SCE’s surveyed customers expressing at least “a little interested” and 16% of customers stating they are “very interested.”¹⁸

Absent export opportunities or participation in grid service programs, customers are likely to use these vehicles solely to meet onsite needs. The CEC’s Roadmap anticipates this outcome, noting that customers can prioritize vehicle-to-home (“V2H”) applications to offset consumption during peak time-of-use periods, *potentially through systems voluntarily islanded from the grid.*¹⁹ While such use cases can provide benefits, including an estimated 5 GW reduction in residential peak load by 2030, **these systems may operate disconnected from the grid and therefore limit broader system value. Systems configured in this manner can mitigate a customer’s impact on the grid but do nothing to mitigate their neighbor’s impact.**

¹⁷ California Energy Commission. *A Roadmap to Unlocking the Benefits of Bidirectional Charging*. March 2026. Page 3.

¹⁸ SCE-04A at p. 125, lines 10-16.

¹⁹ The CEC roadmap refers to these systems as “grid disconnected.”

This trajectory risks pushing customers to maximize their private benefits rather than aligning their behavior with the system's needs. While non-exporting applications of bidirectional EV charging systems may occasionally coincide with grid benefits, they will not consistently deliver optimized system outcomes. Customers acting independently will not necessarily discharge at times of the highest grid value, nor will they support load shaping in ways that reduce overall system costs, including the immense estimated costs of distribution system upgrade costs identified by the Commission's Electrification Impact Studies – Part 1, the utilities' Electrification Impact Studies – Part 2, and Cal Advocates' Distribution Grid Electrification Model.

VGIC is particularly concerned that this dynamic could lead to unintended cost shifts. In studies on the benefits of electrification, increased grid consumption during off-peak times reduces rates for all Californians.²⁰ Shaping increased load, therefore, becomes important to maximize ratepayer savings, but **simply removing significant overall load from the system does not address California's affordability crisis and could lead to cost shifts among customers.**

In the absence of export compensation or clear grid-participation signals, **technology providers may increasingly design products for self-consumption or grid-disconnected use cases.** Early generations of bidirectional EV charging solutions are already marketed primarily for grid-isolated backup power. Technology development cycles are occurring now for products that will be deployed over the next three to five years. Without clear policy direction, these systems are likely to be optimized for customer-only use, including daily cycling for time-of-use arbitrage and elective islanding, even under “blue sky” or normal operating conditions.

²⁰ The Public Advocates Office. Abishek, Benjamin Hoffman, Leo Steinmetz, and Richard Khoe. *Distribution Grid Electrification Model 2025*. <https://www.publicadvocates.cpuc.ca.gov/-/media/cal-advocates-website/files/presentations/251030-public-advocates-office-dgem-2025-slides.pdf>

To avoid this outcome, the Commission should establish a framework that encourages bidirectional EV charging systems to remain connected to, and actively support, the grid. Providing a pathway for export and grid services will help align customer incentives with system needs, unlock the full value of these DERs, and ensure that benefits are shared across all ratepayers.

VGIC reiterates that, in the case of denying export compensation for bidirectional EV charging customers, there is an immense ratepayer cost of inaction, as this misstep will directly lead to greater development and deployment of very large home batteries (in the form of 80kWh/11kW bidirectional EV charging systems) that support grid defection and reduced utility revenue from these customers, directly undermining the Commission's affordability goals.

VI. CONCLUSION.

VGIC appreciates the opportunity to submit this Opening Brief. For the above reasons, the Commission should require SCE to include dynamic export compensation based on unscaled marginal costs as part of its dynamic rate.

Respectfully submitted,
/s/ Zach Woogen
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